

Institute of Actuaries of India

Subject CB2-Business Economics

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u>	B	[1.5 Marks]
<u>Solution 2:</u>	B	[1.5 Marks]
<u>Solution 3:</u>	A	[1.5 Marks]
<u>Solution 4:</u>	B	[1.5 Marks]
<u>Solution 5:</u>	C	[1.5 Marks]
<u>Solution 6:</u>	C	[1.5 Marks]
<u>Solution 7:</u>	A	[1.5 Marks]
<u>Solution 8:</u>	B	[1.5 Marks]
<u>Solution 9:</u>	C	[1.5 Marks]
<u>Solution 10:</u>	D	[1.5 Marks]
<u>Solution 11:</u>	B	[1.5 Marks]
<u>Solution 12:</u>	C	[1.5 Marks]
<u>Solution 13:</u>	D	[1.5 Marks]
<u>Solution 14:</u>	D	[1.5 Marks]
<u>Solution 15:</u>	D	[1.5 Marks]
<u>Solution 16:</u>	A	[1.5 Marks]
<u>Solution 17:</u>	D	[1.5 Marks]
<u>Solution 18:</u>	C	[1.5 Marks]
<u>Solution 19:</u>	B	[1.5 Marks]
<u>Solution 20:</u>	C	[1.5 Marks]
<u>Solution 21:</u>	C	[1.5 Marks]
<u>Solution 22:</u>	A	[1.5 Marks]
<u>Solution 23:</u>	D	[1.5 Marks]
<u>Solution 24:</u>	A	[1.5 Marks]
<u>Solution 25:</u>	C	[1.5 Marks]
<u>Solution 26:</u>	C	[1.5 Marks]
<u>Solution 27:</u>	A	[1.5 Marks]
<u>Solution 28:</u>	B	[1.5 Marks]

Solution 29:

i)

a) Nominal GDP

Goods	2001			2002			2003		
	Qty	Price	(P*Q)	Qty	Price	(P*Q)	Qty	Price	(P*Q)
A	37	3	111	40	4	160	39	5	195
B	15	18	270	17	20	340	18	21	378
C	10	50	500	11	60	660	10	65	650
Nominal GDP			881			1160			1223

[1]

b) Real GDP

Goods	2001			2002			2003		
	Qty	Price	(P*Q)	Qty	Price	(P*Q)	Qty	Price	(P*Q)
A	37	3	111	40	3	120	39	3	117
B	15	18	270	17	18	306	18	18	324
C	10	50	500	11	50	550	10	50	500
Real GDP			881			976			941

[1]

c) GDP Deflator = Nominal GDP/Real GDP *100

Year	Nominal GDP	Real GDP	GDP Deflator
2001	881	881	100.0
2002	1160	976	118.8
2003	1223	941	130.0

[1]

ii) The limitations listed below show why measures of aggregate output and income are unable to reflect the 'true' value of these variables:

GDP and GNI do not include non-marketed output. GDP measures the value of goods and services that are traded in the marketplace and that generate incomes for the factors of production. Yet some output of goods and services is not sold in the market and does not generate any income; this is called 'non-marketed output'. An example is one's own work on repairing and improving one's home; if the home repairs were carried out by hired workers, GDP would be greater by the amount of their wages.

GDP and GNI do not take into account quality improvements in goods and services. The quality of many products improves over time, yet this is not taken into account in calculating the value of total output. Technological advances often permit improved products to be sold at lower prices (for example, mobile phones and computers). This process offers significant benefits to consumers, which do not show up in GDP and GNI figures.

GDP and GNI do not account for the value of negative externalities, such as pollution, toxic wastes and other undesirable by-products of production. Virtually all countries contribute to environmental degradation, reducing society's wellbeing, though this is not reflected in GDP/GNI figures.

GDP and GNI do not take into account the depletion of natural resources. The depletion of natural resources (rainforests, wildlife, agricultural soils, etc.) also reduces society's wellbeing, yet is not taken into consideration.

GDP and GNI and differing domestic price levels. Goods and services often sell for very different prices in different countries. To see the implications of this, suppose there are two hypothetical countries that have an identical GDP per capita of INR 1000. Imagine that both countries produce only one identical good, which sells for INR 100 in country A and for INR 200 in country B. The INR 1000 per person in country A can buy 10 units of the good, but the same amount of money can buy only 5 units of the good in country B. Therefore, the INR 1000 of GDP per person in country A, the low-price country, has far greater purchasing power. Purchasing power refers to the quantity of goods and services that can be bought with money. Although the two countries have identical GDPs per capita in terms of INR, the population of country A can buy double the amount of the good than the population of country B. Clearly, the population of country A on average has a higher standard of living than the population of country B. If international comparisons of GDP do not account for differing price levels across countries, we get a highly misleading picture of standards of living in different countries.

GDP and GNI do not include output sold in underground (parallel) markets.

Here we have the case where goods are traded in markets and do generate incomes, but they go unrecorded and therefore are not included in GDP/GNI. An 'underground market' (also known as a 'parallel' or an 'informal market') exists whenever a buying/selling transaction is unrecorded. It may involve the sale of legal goods and services, such as reselling a good at a higher price if there is a price ceiling; or when a plumber does repairs in your home and does not report the income received to avoid paying taxes or in transactions involving illegal goods and services (such as drugs). In this case as well, estimates of the size of underground markets can be made, and when added to the official (or recorded) economy, can arrive at a closer approximation of 'true' GDP and GNI.

[4]

[7 Marks]

Solution 30:

To achieve an inflation target, central banks use monetary policy. If predicted inflation is higher than the target, they use contractionary policy to increase interest rates and lower aggregate demand, thus lowering the rate of inflation. If predicted inflation is lower than the target, they use expansionary policy to lower interest rates and increase aggregate demand.

Advantages of inflation targeting

A lower rate of inflation. Particularly in countries that have experienced persistent high rates of inflation, inflation targeting can be a method to reduce it.

A more stable rate of inflation. This refers to reduced fluctuations in the rate of inflation, likely to be achieved when a target rate is pursued.

Improved ability of economic decision-makers (firms, consumers) to anticipate the future rate of inflation.

Public knowledge about the central bank's objectives on inflation reduces uncertainty and facilitate economic decision making about the future (such as investment decisions).

Greater co-ordination between monetary and fiscal policy. Knowledge about inflation targets allows the government to plan its fiscal policy to complement the central bank's monetary policy.

Greater central bank transparency and accountability. The central bank becomes more open about its activities and more accountable to the government and the public. If it fails to bring inflation close to the target, it must provide an explanation.

Disadvantages of inflation targeting

Reduced ability of the central bank to pursue other macroeconomic objectives. If the central bank focuses only or mainly on inflation at a particular target rate, it is unable to use monetary policy to pursue other goals, such as for example, the full employment level of real GDP or exchange rate stability.

Reduced ability of the central bank to respond to supply-side shocks. In the event of a supply-side shock, such as a sudden increase in oil prices, leading to cost-push inflation and stagflation, the central bank may need flexibility to pursue an expansionary policy to bring the economy out of recession, and this may mean a higher rate of inflation than the target.

Reduced ability of the central bank to deal with unexpected events, such as financial crises. A financial crisis may also require an expansionary monetary policy, which might also lead to inflation higher than the target.

Finding an appropriate inflation target. An inflation target that is too high or too low can lead to problems. If it is too low, it may lead to higher unemployment; if it is too high, it could lead to the problems resulting from high inflation.

Difficulties of implementation. Inflation targeting is based heavily on forecasts of future inflation and economic activity, and forecasts are often highly unreliable.

[8 Marks]

Solution 31:

The five main factors affecting the demand for money are-

Money national income - An increase in money national income (either arising from an increase in real national income or an increase in prices) will lead to an increase in expenditure on goods and services and hence an increase in the demand for money with which to buy them

The frequency with which people are paid - The less frequently people get paid, the greater the demand for money

Financial innovations, e.g. the use of credit cards reduces the demand for money; the use of cash machines and debit cards and the payment of interest on current accounts increase the desirability of holding money

Speculation regarding the expected returns on financial assets - if returns are expected to fall, there will be an increase in the demand for money; if the domestic currency is expected to appreciate, there will be an increase in the demand for money

The rate of interest – The opportunity cost of holding money as an asset is the interest foregone by not holding other higher-earning assets such as shares and bonds, so the higher the interest rate on assets such as bonds, the greater the opportunity cost of holding money and so the lower the demand for money.

[5 Marks]

Solution 32:

- i) The subsidy per unit is the vertical difference between the two supply curves, or $7 - 5 = \text{INR } 2$ per unit. [1]
- ii) Imports before the subsidy are 9 million units ($= 16 \text{ million} - 7 \text{ million}$).
Imports after the subsidy are 4 million units ($= 16 \text{ million} - 12 \text{ million}$).
Therefore, imports fall by 5 million units ($= 9 \text{ million} - 4 \text{ million}$). [2]
- iii) The price received by domestic producers increases from INR 5 to INR 7, or by INR 2 per unit, which is the subsidy per unit.

The quantity produced also increases from 7 million to 12 million units, or by 5 million units.

Therefore, producer revenue increases from $\text{INR } 5 \times 7 \text{ million units} = \text{INR } 35 \text{ million}$ before the subsidy to $\text{INR } 7 \times 12 \text{ million units} = \text{INR } 84 \text{ million}$ after the subsidy, or by INR 49 million ($= \text{INR } 84 \text{ million} - \text{INR } 35 \text{ million}$).
Producers are better off. [3]
- iv) Domestic Consumers are not affected. They pay the same price, INR 5, and they buy the same quantity, 16 million units before and after the subsidy. The only difference is that now less of the quantity consumed is imported and more is produced domestically. [1]
- v) Exports of foreign producers to the country granting the subsidy fall by an amount equal to the fall in the country's imports, calculated above to be 5 million units.
Export revenues of the foreign producers fall by an amount equal to the fall in the quantity of exports (5 million units) times the world price (INR 5), which is INR 25 million ($= \text{INR } 5 \times 5 \text{ million}$). Therefore, foreign producers are worse off. [2]

[9 Marks]

Solution 33:

- i) Motor Third Party Insurance is a merit good because it satisfies the two main conditions of a merit good:
 - a. Its benefits are not fully appreciated by user as people believe that they will be safe, or they do not understand the risks
 - b. It confers external benefits i.e. it offers protection for other people who might be injured and whose cars might be damaged by the insured driver. [2]
- ii) **Advertising and monopolistic competition:**
 - Advertising is used by monopolistic competition to create product differentiation. Hence, control some market share and charge higher price to customers.

- Advertising a product is intended to shift the product's demand curve to the right. Thus, it creates more awareness among the consumers about the product.
- The products' features are informed to consumers relative to competitors' products
- It makes the product's demand less price-elastic by creating brand loyalty in minds of consumers. This reduces number of substitute goods in minds of consumers.
- At price P_1 , the quantity demanded is Q_1 . After advertising, the demand curve shifts to the right and an increased quantity (Q_2) to be sold at the original price. However, as the advertising has also increased brand loyalty and so reduced the elasticity of demand, the firm may be able to do better than this by increasing its price to P_2 and still increasing sales.

[5]

[7 Marks]**Solution 34:**

i) In order to practise third-degree price discrimination:

- The firm must have some control over the price it sets, *ie* it must face a downward-sloping demand curve.
- It must not be possible for consumers to buy a good and then to resell it on to other consumers at a higher price.
- The elasticity of demand for the good must differ between different groups of consumers.
- It must be possible (and not prohibitively expensive) to identify the different groups of consumers and their different elasticities of demand.
- The characteristic chosen to differentiate between groups of consumers must be relatively easy to observe, acceptable to the consumers and legal.

[4]

ii) Price discrimination is beneficial to the society.

Advantages:

- Government charges differential prices for senior citizens, children below certain age group and adult for bus and railways. This helps the senior citizens and children below certain age to enjoy lower fare.
- Government / Public Institutions charge differential entry fees for foreigners and residents. This would help to earn higher revenue for the institution.
- Different prices are charged for the same product in different locations. This may help to redistribute the wealth among rich and poor people.
- Some institutions offer different versions of the product like television with different prices. This would help people belonging to differential income groups to meet their needs.
- Price discrimination reduces dead weight loss and increases total welfare.

Disadvantages:

- Price discrimination can be used by supplier to increase market power of the firm in the market. It may be harmful to competition. e.g. various discounts offered to customers to increase sale of the product
- Price discrimination declines consumer surplus. It enables a transfer of money from consumers to firms.
- The firm can charge as much the consumer is willing to pay, this may increase profits of the firm as firm can charge higher price than previously set uniform price.

[4]

[8 Marks]

Solution 35:**i) Minimum Efficient Scale-**

- Minimum efficient scale (MES) is the output level beyond which no further economies of scale can be achieved.
- It is the balance point at which a company can produce goods at competitive price.
- Achieving minimum efficient scale minimises long run average total cost (LRAC).
- No further economies of scale can be achieved beyond this scale.

[2]

ii) The major factors which explain the shape of a firm's average cost curve in short run are as follows:

- The law of diminishing returns is the relevant factor in determining short run average costs.
- As increasing amounts of a variable factor of production are added to a given amount of a fixed factor, marginal and average product initially tend to rise implying a fall in short run average costs. After a certain point, there is fall in marginal and average product therefore, short run average cost increases.

[4]

[6 Marks]**Solution 36:****i) The three stages of technological change are as follows:**

- Invention- the development of new ideas and new products
- Innovation- putting new ideas into practice
- Diffusion- the spread of ideas to other organisations which may copy or adapt the original idea

[3]

ii) The Government may intervene in the market to encourage firms to undertake research and development in following ways:

- Public provision- The government could perform its own R&D through research institutions it controls, or by funding universities. This step avoids duplication of research work and saves time.
- R & D subsidies- As an alternative to performing its own research, the government could pay subsidies to firms conducting R&D. This would reduce a firm's disincentive to conduct R&D in the form of both cost and risk.
- Co-operative R&D- The government could encourage a number of firms to work together on a particular piece of R&D. This reduces the danger of duplication of work and helps pooling of scarce resources to achieve the results.
- Diffusion policies – These policies are concerned with:
 - the provision of information on new technologies
 - the encouragement of the use of new technologies
- The patent system- The Government offers a temporary legal monopoly awarded to an inventor who registers an invention. This is because the details of patents are published, but the patent holder is given legal protection in the use of the ideas within the patent for a period of time.
- Other policies- The Government may design other policies such as education and training policies, competition policy and policies on standards and compatibility to promote research and development.

[5]

[8 Marks]
